Supplement S1: review guide

Inference Delay and Deadline Handling in Thermal Controller Evaluation: A Retrospective Simulation Study

Read this guide together with the 14-page manuscript and the separate Supplement_S1 ZIP. The ZIP contains the saved inputs, implementation details, SHA-256 manifest, and executable scripts. This PDF provides navigation and the two supplementary command figures.

Run the saved-result checks

Extract the ZIP, install the requirements if necessary, then run:

```
python -B scripts/reproduce_review.py
```

Python 3.10+, NumPy, and Matplotlib are required. The command checks the manifest, paired results, 57 replay/fallback contrasts, physical components, alpha selection, and command increments, and writes six main figures plus two supplementary figures to outputs/. No model, GPU, or plant simulator is invoked.

Quick routes to the principal results

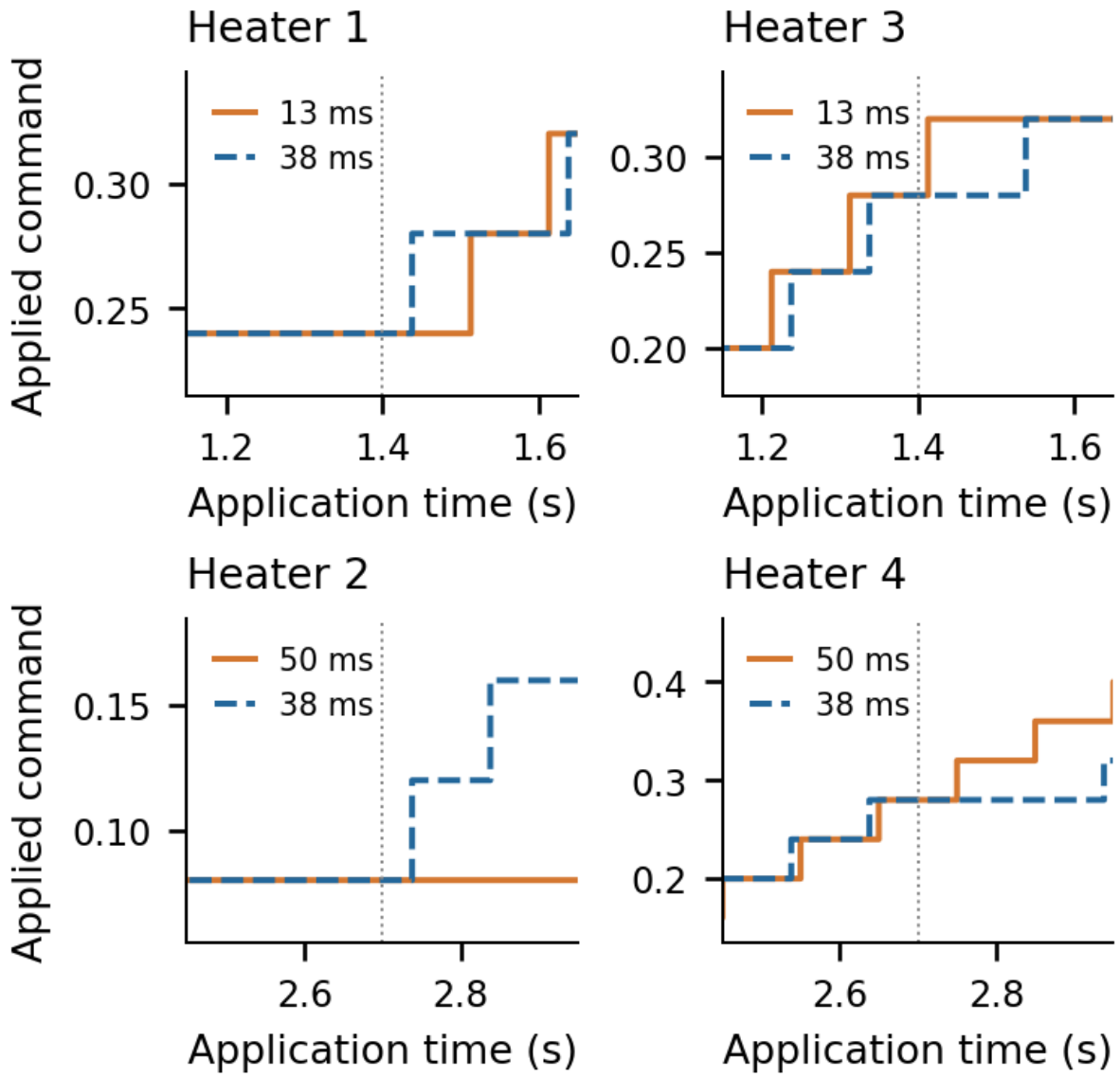
Manuscript	Saved input / script	Generated output
Tables 7 and 8	data/primary/episode_metrics.csv scripts/reproduce.py	table6_primary_costs.csv table7_paired_comparisons.csv
Table 9 and Figure 4	data/replay/episode_metrics.csv scripts/reproduce_replay.py scripts/make_replay_figure.py	replay_paired_results.json figures/replay_fallback_sensitivity.pdf
Table 10	data/revision/cost_components.csv scripts/reproduce_physical.py	physical_checks.json Subtract the reported condition means.
Table 11	data/numerical/extended_metrics.csv scripts/reproduce.py	table8_numerical_sensitivity.csv
Figure S2	data/minor_revision_r2/deadline_commands.csv scripts/make_deadline_figure.py	figures/deadline_command_response.pdf

Paths in the last column are under outputs/. Some output filenames retain historical table numbers. TABLE_FIGURE_MAP.md covers all 11 main tables and six main figures. METHODS.md provides residual settings (Section 2), PINO settings (Section 3), injected delays (Section 6), and N=600 normalization (Section 7).

Scope of reproduction

The package supports saved-result reanalysis and figure regeneration. It omits model weights, full training inputs, raw timing journals, and full quadrature/field archives. It cannot independently execute new closed-loop conditions or repeat the complete raw-input audit. Original, numerical-validation, and new replay/fallback records are distinguished in README.md.

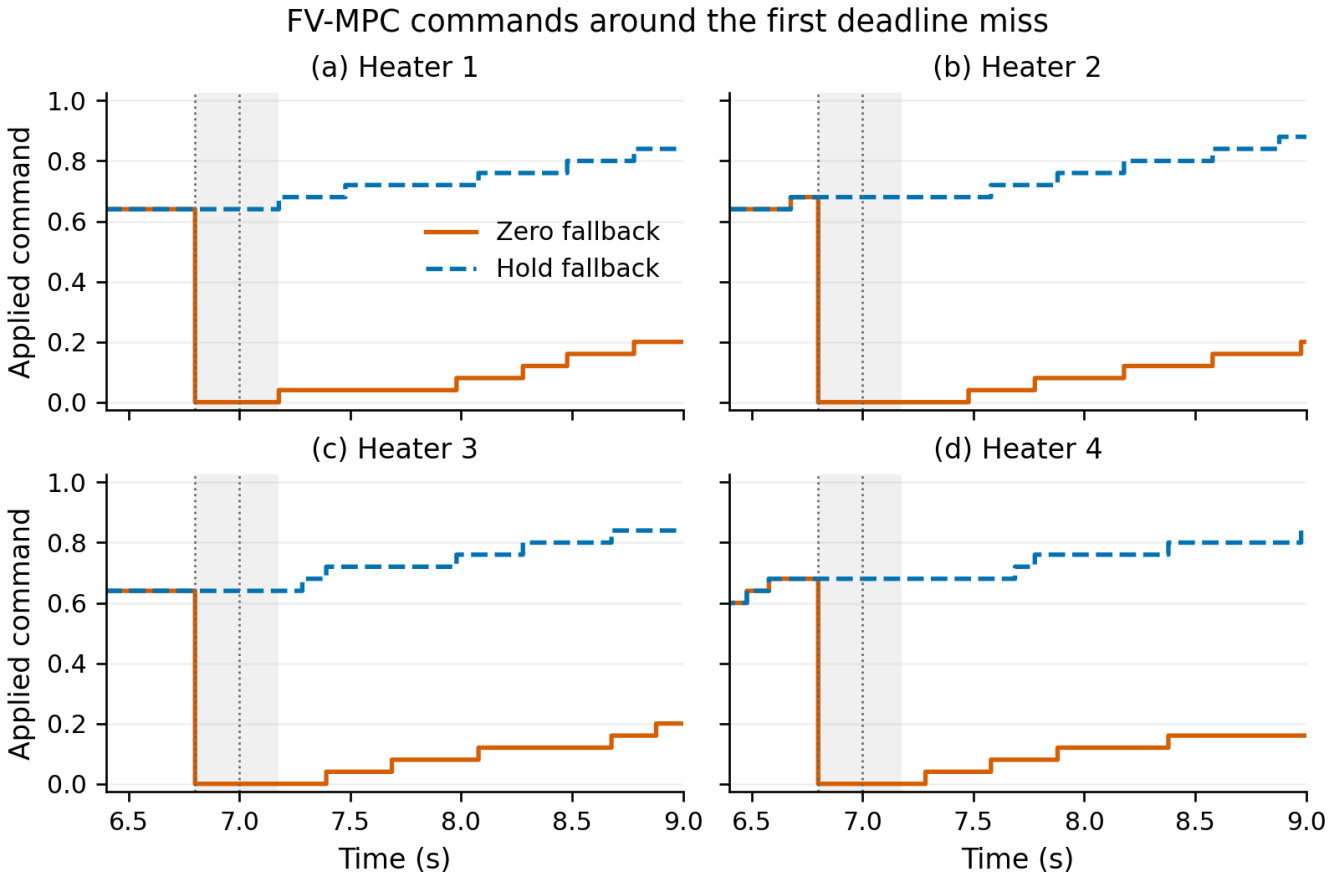
Figure S1. Selected C command diagnostic



Archived command comparison for the selected C diagnostic. The detailed case protocol, first differing requests, individual command vectors, and score margins are retained in methods/selected_C_detail.tex. Inputs are in data/selected_C/; scripts/plot_results.py generates this figure as part of scripts/reproduce.py.

C uses a different frozen direct scorer from D. Its episode was selected post hoc. It supports a conditional implementation diagnostic and does not estimate population performance or identify an optimal physical delay. The main manuscript reports that recomputed closed-loop decisions remain sensitive to the plant integration step.

Figure S2. First deadline miss in D



Applied commands for the four heaters of episode I2_240070 under original interleaved replay at a 10-ms plant cap. Solid orange: zero fallback. Dashed blue: hold fallback. Dotted lines mark expirations at 6.8 and 7.0 s; shading ends at the next nonfallback application (7.177863943 s). Curves use actual application times with right-continuous holds.

Immediately before the first expiration, the command is [0.64, 0.68, 0.64, 0.68]. Zero fallback resets it to [0, 0, 0, 0]; hold retains it. The next nonfallback commands are [0.04, 0, 0, 0] and [0.68, 0.68, 0.64, 0.68], respectively. Subsequent commands reflect each condition's evolving state. This shows the intervention and subsequent feedback, without establishing a safe hardware policy.

The episode is unchanged from main Figure 5 and was selected by zero-delay FV cost before inspecting hold performance. The two raw case seals and identical request/deadline schedules were checked. Source hashes and audit scope are in data/minor_revision_r2/deadline_command_evidence.json.

Why N=600 is not an event count

Both cases contain 592 requests and 592 application/deadline records. The fixed denominator is still the 600 nominal slots in 60 s. Unchanged commands contribute zero to the increment sum. Reconstructing the penalty from all saved commands gives 0.00001872 for zero and 0.000002666666666666667 for hold, matching the stored metrics.